WORKSHEET - NAPLAN Year 9

Number and Algebra

Fractions, Decimals and Percentages (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-72097

Penny has four 20-cent pieces.

Fi has three 20-cent pieces.

Sophie has four 20-cent pieces.

How much money do they have in total?

- $\$1.10$
- $\$2.00$
- $\$2.20$
- $\$2.60$

2. Number and Algebra, NAP-66182

Andreas has $8 to buy batteries for his toy racing car.

Each battery costs $1.60 and he buys 4 batteries.

Which expression shows how much money he has left?

- $\$8 - \$1.60$
- $\$8 - (4 \times \$1.60)$
- $\$8 - \$1.60 + \$1.60 + \$1.60 + \$1.60$
- $4 \times (\$8 - \$1.60)$

3. Number and Algebra, NAP-89702

500 kilograms of a concrete material is made by mixing 200 kilograms of cement with 300 kilograms of sand.

What percentage of the concrete is made up of sand?

- 0.05%
- 0.3%
- 0.6%
- 3%
- 30%
- 60%

4. Number and Algebra, NAP-83793

Charlie has $20 to buy a can of soda and a chocolate bar from the shop.

$2.80 $1.50

How much change should Charlie be given?

- $\$15.70$
- $\$16.70$
- $\$17.10$
- 24.30

5. Number and Algebra, NAP-36746

Sharnie is buying eggs for her restaurant's kitchen.

Each carton contains 12 eggs.

If she needs 90 eggs, how many cartons will she need to buy?

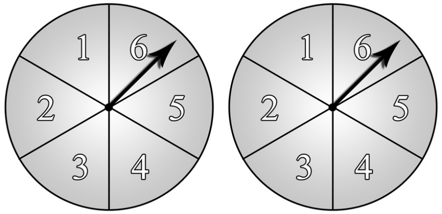
6. ☒ Number and Algebra, NAP-83823

Which of these is the closest to 1?

0.9 ○ 0.99 ○ 1.01 ○ 1.001 ○

7. ☒ Statistics and Probability, NAP-71836

Jon spins each pointer 50 times.



Each time he added the numbers that the pointers landed on.

His results are shown below.

Sum of numbers	Number of spins
2	1
3	2
4	3
5	6
6	8
7	9
8	7
9	5
10	4
11	4
12	1
Total	50

What percentage of the spins resulted in a sum of 9?

☐ %

8. ☒ Number and Algebra, NAP-25147

When 1 mm of rain falls on 1 m^2 of a pool, 1 litre of water is collected.

A pool of what surface area is needed to collect 6000 litres from a rainfall of 15 mm?

☐ m^2

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Worked Solutions

Worked Solutions

1. Number and Algebra, NAP-72097

$\$2.20$

2. Number and Algebra, NAP-66182

$\$8 - (4 \times \$1.60)$

3. Number and Algebra, NAP-89702

$$\begin{aligned}\text{Percentage} &= \frac{300}{500} \times 100\\ &= \frac{3}{5} \times 100\\ &= 60\%\end{aligned}$$

4. Number and Algebra, NAP-83793

$$\begin{aligned}\text{Change} &= 20 - (2.80 + 1.50)\\ &= 20 - 4.30\\ &= \$15.70\end{aligned}$$

5. Number and Algebra, NAP-36746

$90 \div 12 = 7$ remainder 6

$\therefore$ Sharnie need 8 cartons.

6. Number and Algebra, NAP-83823

1.001

7. Statistics and Probability, NAP-71836

$$\begin{aligned}\text{Percentage} &= \frac{\text{number of 9's}}{\text{number of spins}} \times 100\\ &= \frac{5}{50} \times 100\\ &= 10\%\end{aligned}$$

8. Number and Algebra, NAP-25147

If 15 mm of rain falls 1 m^2

$\Rightarrow$ 15 litres collected

$\therefore$ Surface Area of pool needed

$$\begin{aligned}&= \frac{6000}{15}\\ &= 400\ m^2\end{aligned}$$

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